

Mateo Alado

(510) 543-1444 | aladomateo@gmail.com | linkedin.com/in/mateoalado | github.com/matesuu

PROFESSIONAL EXPERIENCE

Software Engineer Intern (Applied AI)

June 2026 – August 2026

Lawrence Berkeley National Laboratory

Berkeley, CA

- Engineered a pipeline of 4 HPC-enabled agents to automate scientific literature ingestion, extraction, and context retrieval for over 16,000 Advanced Light Source (ALS) publications
- Reduced context retrieval and inference time from over 3 minutes to under 5 seconds by designing a custom PostgreSQL-backed GraphQL API for low-latency queries
- Designed a real-time graph rendering UI with React expanding information access and usability for 450+ Berkeley Lab users across ALS and NERSC

Applied Machine Learning Researcher

January 2026 – June 2026

California State University, East Bay

Hayward, CA

- Developed a backend inference engine using PolyG, reducing average LLM token usage by 79.7% compared to similar open-source RAG frameworks
- Benchmarked against GraphRAG and reasoning-based baselines, achieving the highest F1 score (0.71) and 89% hit rate when compared to Microsoft GraphRAG and Graph-CoT

PROJECTS

Multi-Model Coding Agent | *TypeScript, Bun, Perplexity API, Cloudflare Workers, Cloudflare D1*

- Developed a terminal-native AI coding agent with MCP tooling leveraging Perplexity's deep reasoning models for context-aware refactoring and debugging for developers
- Achieved an 88/100 on SWE benchmarks in multi-file debugging, async concurrency, and SQLAlchemy migrations, outperforming Codex (72) and Qwen 397B (58)

GraphRAG Research QA Application | *Python, TypeScript, React, OpenAI API, FastAPI, Neo4j, Supabase, Docker*

- Built and deployed a full-stack research platform using FastAPI and Supabase for a team of 10 faculty researchers to query and manage 30+ academic papers and datasets
- Architected a RAG-based inference engine using external knowledge graphs for adaptive graph traversal, optimizing long-context prompting and reducing model hallucinations

Visual Speech Classifier | *Python, PyTorch, NumPy, CNN, MediaPipe, OpenCV*

- Engineered a real-time lip-reading classifier and pipeline using MediaPipe Face Mesh and OpenCV webcam capture to classify speech patterns for hearing-impaired individuals
- Achieved a 94% validation accuracy across 24 speech classes through iterative CNN optimization and refinement

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, C/C++, Go

AI/ML: LLMs, RAG, GraphRAG, PyTorch, TensorFlow, LangChain, LangGraph, Agentic AI

Backend: FastAPI, REST APIs, Node.js, Distributed Systems

Database: PostgreSQL, MongoDB, Neo4j, Supabase, Vector Databases

Developer Tools: Docker, Hugging Face, AWS, Git, Linux, Ollama, Cursor, Claude Code

EDUCATION

California State University, East Bay

Hayward, CA

Bachelor of Science in Computer Science

Aug. 2024 – June 2026

- Relevant Coursework: Data Structures and Algorithms, Operating Systems, Artificial Intelligence, Machine Learning, Natural Language Processing (NLP), Software Engineering, Linear Algebra, Statistics and Probability
- Distinctions: 4x Dean's Honor List, HackHayward 2026 Winner, Google Developer Student Club Member

Chabot College

Hayward, CA

Associate of Science in Computer Science

Aug. 2022 – June 2024